CS6023 – GPU Programming
Assignment - 1: Matrix Blur Using CUDA

1 December 2025

1 Problem Statement

You are given a matrix of size $N \times M$ containing integer values. Your task is to compute a blurred output matrix where each cell is replaced by the **ceiling of the average** of itself and all valid neighbouring cells (up to 8 neighbours).

Given an input matrix of size $N \times M$, compute an output matrix of the same size where each entry (i, j) is replaced by:

$$\text{output}[i][j] = \left\lceil \frac{\sum \_\text{all valid neighbours including } (i,j)}{\text{count of valid neighbours} + 1} \right\rceil$$

Each cell may have between 3 and 8 valid contributing neighbours depending on its position:

- Corner cells: 3 neighbours
- Edge cells: 5 neighbours
- Interior cells: 8 neighbours

Blur Operation Illustration

For any cell at (i, j), the neighbourhood consists of coordinates:

$$(i-1, j-1), (i-1, j), (i-1, j+1)$$
$$(i, j-1), (i, j), (i, j+1)$$
$$(i+1, j-1), (i+1, j), (i+1, j+1)$$

Only the valid positions (inside matrix bounds) are included in the average.

2 Input Format

The input file follows this naming convention:

`inputX.txt`

The content of the file is:

- First line: Two integers N and M
- Next N lines: Each contains M space-separated integers representing the matrix

Example:

```
5 8
12 4 7 9 3 2 1 5
8  6 4 2 9 1 3 0
5  5 5 5 5 5 5 5
1  2 3 4 5 6 7 8
9  9 9 9 9 9 9 9
```

3 Output Format

The output must be written to a file with the corresponding name: `outputX.txt`

Output format:

- First line: Two integers N and M
- Next N lines: The blurred matrix
- Next line: Kernel runtime in seconds

4 Sample Test Case

Input (input1.txt)

```
3 3
1 2 3
4 5 6
7 8 9
```

Output (output1.txt)

```
3 3
3 4 4
5 5 6
6 7 7
0.0001523
```

Explanation

Let us compute the blurred value for the cell at $(0, 0)$:
Valid neighbours: 1, 2, 4, 5
Sum $= 1 + 2 + 4 + 5 = 12$
Count $= 4$
Average $= 12/4 = 3$
Ceiling $\lceil 3 \rceil = 3$
This matches the output at position $(0, 0)$.
Every other cell is computed similarly.

5 Files Provided & Testing Instructions

Each student will download a ZIP folder that contains the following structure:

ZIP Folder Structure

```
Assignment1/
  template.cu
  Inputs/
       input1.txt
       input2.txt
       ...
  Outputs/
       output1.txt
       output2.txt
```

```
    ...
test.sh
```

template.cu This file contains:

- Boilerplate code for reading the input matrix.
- Boilerplate code for writing the output.

Students must **not modify any other part of the template**.

- Compile your `RollNumber.cu` file using `nvcc`.
- runs "*bash test.sh ./ROLLNUMBER.out*" . this will run your program on **every input file** inside the `Inputs/` folder.
- For each input file `inputX.txt`, the script places your output in:

`Program-outputs/outputX.txt`

- The script then compares each generated file against the corresponding official output in the `Outputs/` folder using:

6 Constraints and Notes

- $1 \leq N, M \leq 5000$
- $1 \leq Matrix[i] \leq 50000$
- All computation must be done on the GPU.
- Only the kernel runtime (not file I/O) should be measured.
- Do not change the required output format.
- Do not modify any provided code outside the designated blocks.
- Each student must submit exactly one file: `RollNumber.cu`